Website Development – Complete Class Material

Topic: Introduction to Websites & Basic Website Structure

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1. Introduction to Websites

What is a Website?

A website is a collection of web pages available on the internet under a single domain name.

Examples:

- Google
- YouTube
- Amazon

A website can contain:

- Text
- Images
- Videos
- Forms
- Buttons
- Animations
- Databases
- User Login Systems

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2. Types of Websites

Static Website

- Fixed content
- Simple HTML/CSS
- No database
- Fast loading

Example:

- Portfolio website
- Company profile website

Dynamic Website

- Content changes dynamically
- Uses backend and database

Examples:

- Facebook
- Instagram
- E-commerce websites

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3. How Websites Work

Flow of a Website

```
User → Browser → Internet → Web Server → Website Response
```

Example:

1. User types a website URL
2. Browser sends request
3. Server processes request
4. Website page appears

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4. Important Terms

Term	Meaning
Domain	Website name
Hosting	Place where website files are stored
Browser	Software to open websites
URL	Website address
Server	Computer that serves websites
Frontend	Visible part of website
Backend	Logic and database side

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5. Website Basic Structure

A basic website mainly contains:

```
Website
│
├── Header
├── Navigation Menu
├── Main Content
├── Sidebar (Optional)
└── Footer
```

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6. Main Parts of a Website

1. Header

Top section of website.

Contains:

- Logo
- Website name
- Navigation links

Example:

```
<header>
    <h1>My Website</h1>
</header>
```

---

2. Navigation Bar

Used for moving between pages.

Example:

```
<nav>
    <a href="#">Home</a>
```

```
    <a href="#">About</a>
    <a href="#">Contact</a>
</nav>
```

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3. Main Content Section

Contains actual page information.

Example:

```
<section>
    <h2>Welcome</h2>
    <p>This is my website.</p>
</section>
```

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4. Sidebar (Optional)

Used for:

- Ads
- Recent posts
- Categories

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5. Footer

Bottom section of website.

Contains:

- Copyright
- Contact details
- Social media links

Example:

```
<footer>
    <p>© 2026 My Website</p>
</footer>
```

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7. Technologies Used in Website Development

Technology	Purpose
HTML	Structure
CSS	Design
JavaScript	Interactivity
Bootstrap	Responsive Design
React	Frontend Framework
Django	Backend Framework
MySQL/PostgreSQL	Database

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8. Frontend vs Backend

Frontend

Visible part users interact with.

Technologies:

- HTML
- CSS
- JavaScript

Backend

Handles:

- Database
- Authentication
- Business logic

Technologies:

- Python
- Django
- Node.js
- PHP

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9. Basic HTML Structure

```
<!DOCTYPE html>
<html>
<head>
    <title>My Website</title>
</head>

<body>

    <header>
        <h1>My Website</h1>
    </header>

    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Contact</a>
    </nav>

    <section>
        <h2>Welcome</h2>
        <p>This is the content area.</p>
    </section>

    <footer>
        <p>Copyright 2026</p>
    </footer>

</body>
</html>
```

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10. Steps to Create a Website

Step 1 – Create Project Folder

Example:

```
website-project
```

Step 2 – Create HTML File

```
index.html
```

Step 3 – Open in Browser

Step 4 – Add CSS

Step 5 – Add JavaScript

Step 6 – Host Website

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11. Popular Website Builders

Tool	Usage
Wix	Drag & Drop Website Builder
WordPress	Blogging & CMS
Shopify	E-commerce
Webflow	Visual Web Design

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12. Website Hosting Platforms

Platform	Purpose
GitHub Pages	Free static hosting
Netlify	Frontend hosting
Vercel	React/Next.js hosting
PythonAnywhere	Python hosting

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13. Advantages of Websites

- Online presence
- Brand awareness

- 24/7 availability
- Marketing
- Customer interaction
- Online sales

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14. Real-World Website Categories

Category	Example
E-commerce	Amazon
Social Media	Facebook
Educational	Coursera
News	BBC
Portfolio	Personal websites
CRM	Business management websites

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15. Class Activity

Activity 1

Create a simple webpage with:

- Header
- Navigation
- Paragraph
- Footer

Activity 2

Add:

- Background color
- Images
- Buttons

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16. Interview Questions

Basic Questions

1. What is a website?
2. Difference between static and dynamic websites?
3. What is HTML?
4. What is the role of CSS?
5. What is frontend?
6. What is backend?
7. What is hosting?
8. What is a domain?
9. Explain website structure.
10. Difference between website and web application?

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17. Practical Task

Task

Create a simple company website containing:

- Logo
- Navigation bar
- About section
- Services section
- Contact section
- Footer

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18. Conclusion

A website is the foundation of modern digital presence. Understanding website structure and basic technologies is the first step toward becoming a web developer.

Students should next learn: 1. HTML 2. CSS 3. JavaScript 4. Responsive Design 5. Backend Development 6. Database Integration